

A Visible Light-Based Optical Camera Communication System for Text Data Transmission

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Abstract—The technology of Optical Camera Communication (OCC) has been gaining significant attention within the Visible Light Communication (VLC) systems community. This technology employs a Light-Emitting Diode (LED) as a transmitter and a camera as a receiver to establish a communication link. Data is encrypted and conveyed by modulating the LED, which is subsequently decoded by a camera. This promising technology is suitable for both indoor and outdoor applications. In this paper, we address the problem of optimizing OCC systems for improved data transmission. Our primary aim is to propose a rapid, robust, and practical OCC system that utilizes an LED array to enhance the emitter's rate for data transfer using On-Off Keying (OOK) modulation. The system leverages a laptop's default camera as an OCC receiver and employs the OpenCV platform for decoding and processing the received data. Our methodology involves evaluating key performance metrics, such as the relationship between the number of LEDs in the array and transmission power, Bit Error Rate (BER), data transmission accuracy, and the exposure time required. The experimental results demonstrate successful text data transmission and reception using a rapid OCC system with a frame rate of 25 fps over a communication distance of 1.1 meters. Conclusively, the proposed system achieves a transmission rate of 32 Kbps under the specified system conditions, showcasing its effectiveness and practical utility in OCC applications.

Keywords—visible light communications, optical camera communication, Light-Emitting Diode (LED) transmitter, CMOS sensor, OpenCV library

I. INTRODUCTION

In recent times, there has been an outstanding surge in the interest in researching Optical Wireless Communication (OWC) technologies, owing to their remarkable characteristics. OWC, which relies on wireless

communication via the optical spectrum, has emerged as an attractive complementary technology to radio frequency-based wireless technologies for future communication networks, encompassing fifth and sixth-generation communication systems. Distinguished features of OWC technologies comprise a broad spectrum, high data rate, minimal latency, superior security, economical cost, and low power consumption [1]. OWC technology encompasses the whole optical spectrum including ultraviolet, visible, and infrared wavelengths, as shown in Fig. 1. This provides a wide theoretical bandwidth of wavelengths which is much greater than the 300 GHz bandwidth of RF technology. OWC includes different transmission systems for both indoor and outdoor ambiances encompassing Free Space Optics (FSO), Visible Light Communications (VLC), LiFi, and Optical Camera Communication (OCC) [2]. The VLC technology, which is a subgroup of OWC as realized in the IEEE 802.15.7 standard, was originally designed in 2000 to cater to the demands of mobile networks with high data capacity and to supplement RF-based mobile communications [3]. The growing publicity of VLC expansion can be attributed to its ability to efficiently utilize existing lighting infrastructure based on Light-Emitting Diodes (LEDs) in the role of a transmitter, thereby offering grandly higher radiant effectiveness, information connections, and indoor localization [4, 5]. An optical camera communication is a VLC system that employs image sensors and a camera for data transmission and the comparison between visible light communication and optical camera communication is presented in Table 1 [6]. OCC finds its utility in a variety of applications, including indoor positioning, mobile automaton navigation, and provision of tiny metainformation. There exists an extensive array of prospective uses for OCC in the realm of the Internet of

Things (IoT), encompassing augmented actuality, appliance-to-appliance communications, digital signage, exhibition serving, airline service, indoor positioning, vehicular communications, and underwater communications [7].

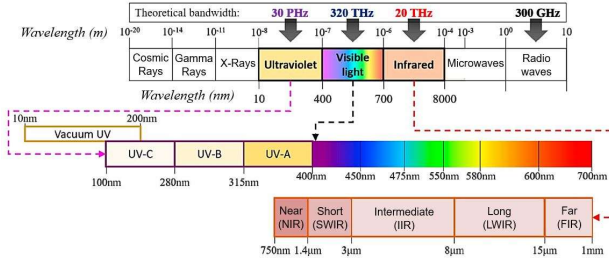


Fig. 1. The electromagnetic spectrum.

TABLE I. VLC AND OCC COMPARISON

Characteristics	Optical Camera Communication	Visible Light Communication
Transmission Distance	The transmission distance is up to several kilometers.	The transmission distance is low.
Wavelength	OCC uses UV, IR, and Visible light.	The wavelength used is Visible light.
Receiver	The receiver type for OCC is the camera.	The receiver type for VLC is the photodetector.
Data Rate	The data rate is lower than VLC.	The data rate is high.

Lately, cameras have become omnipresent gadgets that are effortlessly incorporated into each smart gadget, such as smartphones and laptops. They are therefore regarded as an optimal wireless communication platform that leverages an unlicensed spectrum, resulting in minimal installation costs and energy consumption. However, due to the development of smart devices used now, and the plethora of operating systems available, including but not limited to Android, Windows, and iOS, it is important to note that the OCC application cannot be universally utilized across all device types. OCC has emerged as a highly promising technology for employment in a diverse range of settings, including both indoor and outdoor applications [8, 9]. The camera as receiver encompasses two distinct varieties: global shutter and rolling shutter as shown in Fig. 2. Global shutter cameras capture the entire frame at once, thereby capturing either the ON or OFF state of an LED. Conversely, rolling shutter cameras capture the image through row-by-row exposure. For a set frame rate, the camera exposure time must be configured to accommodate the fluctuating ON-OFF status of the LED and successfully capture the black-and-white striped image captured by the Complementary Metal-Oxide-Semiconductor (CMOS) sensor [10].

The reliance on conventional Radio-Frequency (RF) communication becomes a critical concern in scenarios where it is either impractical or ineffective due to various factors such as high interference, background noise, bandwidth saturation, underwater environments, or even intentional signal jamming. In such situations, Optical

Camera Communication (OCC) emerges as a popular alternative for short-range wireless communication, particularly in line-of-sight conditions. The advantages of OCC include its lightweight visual optical channels, which are relatively easy and cost-effective. However, the data rates achieved by conventional OCC systems, which typically utilize cameras to catch modulated LED senders, are relatively low, ranging from 80 to 200 bits per second (bps). To achieve higher data rates, more advanced high-speed cameras or complex transmitter and receiver systems are required, which may demand significant computational power or dedicated physical infrastructure. Additionally, shorter exposure times in high-speed cameras necessitate brighter LED transmitters, leading to higher power requirements.

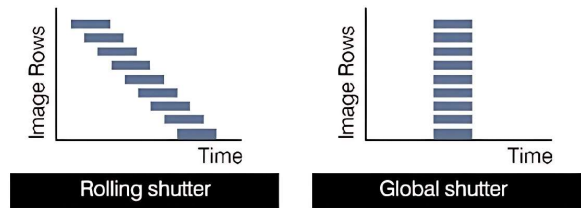


Fig. 2. Schematic of rolling and global shutter.

To overcome these challenges, this paper aims to introduce an experimental system that employs optical camera communication technology for transmitting text data using modulated streaming LEDs and receiving data through a laptop camera. It seeks to use an on-off keying (OOK) modulation which is the simplest amplitude modulation scheme to achieve an efficient data transfer using the LED Array. The main contributions of this paper are presented as follows:

- An experimental system is proposed that utilizes optical camera communication technology for efficient data transmission.
- A modulated streaming RoHs LED is employed for text data transmission, enhancing the capabilities of OCC systems using OOK modulation.
- An easier setup is provided for unidirectional communication in OCC systems.

The rest of this paper is structured as follows: Section II discusses the related work. In Section III, the methods and materials are presented. Section IV provides the system model of the proposed system. Section V discusses the experimental setup. Section VI provides the results. In Section VII, the discussion of the paper is presented. Finally, Section VIII provides the conclusion.

II. RELATED WORK

Optical Camera Communication (OCC) systems have gained significant attention in recent years due to their potential for high-speed data transmission using traditional On-Off Keying (OOK) modulation. OOK modulation is a simple and efficient method for encoding data in optical communication systems, making it a popular choice for OCC applications. The performance of OCC systems is influenced by various factors, including exposure effects,

synchronization, and decoding processes. To mitigate these effects, researchers have proposed various schemes and techniques to enhance the performance of OCC systems. Cahyadi *et al.* [11] proposed an asynchronous On-Off Keying (OOK) modulation scheme to address the synchronization challenges associated with traditional OOK. By combining Manchester coding and the camera rolling shutter effect with OOK, Achari *et al.* [12] demonstrate improved performance. Additionally, Teli *et al.* [13] introduce a rolling-shutter-based Multiple-Input Multiple-Output (MIMO) system utilizing an LED array, promising flicker-free transmission despite a shorter communication distance and limited rotation detection capability. Building upon these advancements, Nguyen *et al.* [14] presents OCC systems employing the MIMO technique and OOK modulation, achieving committed results with a high data rate of 1.920 kbps using an 8×8 LED array. Shiraki *et al.* [15] explore the performance of an Optical Camera Communication (OCC) system employing the traditional On-Off Keying (OOK) modulation scheme. The focus is on demodulation techniques that do not rely on synchronization devices or training sequences, addressing the challenges posed by synchronization and background noise in real-world scenarios. The authors investigate the characteristics of an OCC system when there are slight discrepancies between the periods of reception and transmission. They propose a demodulation procedure based on a maximum-likelihood sequence detection method, which can be efficiently implemented using the Viterbi algorithm. This approach aims to overcome the synchronization issues and mitigate the impact of background noise, thereby enhancing the overall performance and robustness of the OCC system. Shiraki *et al.* [16] focused on an Optical Camera Communication (OCC) system that employs On-Off Keying (OOK) modulation, highlighting its simplicity and effectiveness in facilitating communication. The authors develop a model for the light signal received by a camera, which enables the spatial separation of multiple transmitters through spatial-division multiplexing. This model allows for efficient utilization of the available resources and enhances the system's capacity. Furthermore, the paper investigates the characteristics of an OCC system when there are slight discrepancies between the periods of reception and transmission. It proposes a received light signal model and describes the cost functions that need to be minimized for successful demodulation of the transmitted bit sequences. By optimizing these cost functions, the system can effectively recover the transmitted data, ensuring reliable communication.

This paper proposes an OCC system using OOK modulation. The proposed system leverages the advantages of OCC to enable efficient data transmission. The system is designed to be cost-effective, scalable, and suitable for deployment in areas with limited infrastructure. The details of the proposed system are presented in the following sections.

III. METHODS AND MATERIALS

This section illustrates the Optical Camera Communication (OCC) hardware system employs a comprehensive white RoHS LED configuration, encompassing both emitter and receiver components. The communication channel established between these two ends is characterized as a Line-of-Sight (LOS) connection.

A. Transmitter-RoHS LED Configuration

Light sources play a crucial role in various applications, ranging from illumination to communication and sensing. Among the diverse range of light sources available, Light-Emitting Diodes (LEDs) and lasers stand out as two prominent technologies with unique characteristics and advantages. Table II illustrates the characteristics of the LED and laser.

TABLE II. LED VERSUS LASER

Parameter	LED	Laser
Modulation Speed	Slower	Faster
Safety	Safer	Requires careful handling
Beam Divergence	Wider	Narrower
Spectral Width	Broader	Narrower
Deployment Flexibility	More versatile	More specialized

In this paper, LEDs are the best choice and are more compatible with our system. They offer more flexibility in deployment, as they can be easily integrated into various devices and systems.

LEDs in VLC systems can fulfill two functions, namely providing illumination and enabling communication. The emitted light power from a Light Emitting Diode plays a crucial role in wireless transmission. Nonetheless, the majority of white LED datasheets solely furnish luminous flux in lumens, which is a valuable unit for illumination planning [17, 18]. Consequently, assessments are executed to ascertain the emitted light power in watts for the specified light source, and the emitted light power of an LED can be computed using the following formula:

$$P = \int p(\lambda) d\lambda \quad [W] \quad (1)$$

where P symbolizes the aggregate energy emitted by the LED. and $p(\lambda)$ denotes the source power spectral distribution [19, 20].

The efficiency of transforming optical power from LED into luminosity is described by an optical luminous efficiency as given by Eq. (2).

$$\eta = \frac{\Phi_{lum}}{P} \quad [lm/W] \quad (2)$$

where Φ_{lum} indicates the luminous flux and P is the total power emitted from an LED.

The emitter system presented in this paper employs a controllable array of RoHS-compliant white LEDs, which are meticulously synchronized by a microcontroller to rapidly transition between different shades of white. This innovative technique ensures a seamless and low flicker illumination, achieving a balanced white light appearance

that is imperceptible to the human eye [21, 22]. This versatile approach allows for the encoding of messages through rapid fluctuations in white light intensity, where each sequence represents a unique data transmission. The utilization of RoHS-compliant white LEDs not only ensures environmental sustainability but also provides a reliable and efficient means of optical communication, making this emitter system a promising candidate for various applications requiring high-speed data transfer and energy efficiency.

We used a 3×3 LED array of the RoHS type offers several advantages, including enhanced data transmission rates and improved signal robustness. The LED array is arranged in a square pattern, with three LEDs in each row and column. This configuration provides a uniform distribution of light, which is beneficial for optical camera communication. The array is designed to operate as a single unit, with each LED contributing to the overall light output. Detailed characteristics of the RoHS LED are listed in Table III.

TABLE III. THE ROHS LED SPECIFICATIONS

Characteristic	Value/Description
Wavelength	450–650 nm (visible light spectrum).
Luminous Intensity	100–1000 mcd (millicandela).
Viewing Angle	120–160 degrees (wide viewing angle).
Efficiency	60–80 lumens per watt (lm/W).
Lifespan	50,000–100,000 hours (long-lasting).

B. Receiver-Laptop Camera

Current intelligent technologies like smartphones and laptops possess remarkable capabilities in detecting and processing information. For instance, a CMOS camera sensor was employed to capture data emanating from visible LED light sources. In CMOS sensors, individual pixels are equipped with dedicated conversion circuits to produce digital bits as output. This intricate design, however, results in a reduction in the surface area allocated for light capture, which consequently leads to lower image uniformity [23]. On the other hand, the conveyance of picture information is completed in an extremely parallelized way, which encourages quick imaging and on-chip picture handling abilities. In this paper, we use the default camera of an HP laptop to detect the illuminating light emanating from the transmitter. The camera employs a sequential scanning technique across its pixel matrix, capturing and assembling individual pixel states to construct a complete frame within a time interval denoted as t_{frame} . This comprehensive frame, representing the camera's field of view, exhibits dimensions proportional to the camera's resolution, encompassing both height and width. During this systematic scanning process, the camera adeptly perceives and records the dynamic variations in light states, enabling the accurate interpretation of the visual scene.

Exposure is one of the key characteristics of the OCC system's performance where exposure refers to the

quantity of light recorded by the camera, which is contingent upon the luminous intensity affecting the image plane (known as the image irradiance) and the interval of the exposure is expressed by Eq. (3).

$$\mathcal{E} = Et \quad (3)$$

where E is the power per square meter of image irradiance.

IV. SYSTEM DESIGN

The OCC system is composed of LEDs and image sensors that function as transmitters and receivers, respectively. LED technology is widely utilized in various light-based applications, such as indoor illumination, street signals, and car lights. The rapid switching of LEDs between the ON and OFF states enables high-frequency modulation of LED light [24]. Fig. 3 provides a basic block diagram representation of the OCC system used. It consists of a programmable board that is fed text data controls the speed of transmission using short delay times and helps avoid light flickering significantly. Encoded data is utilized to modulate light at frequencies higher than those detectable by a camera. The LED driver circuit performs all necessary tasks to control the LED array. Ultimately, the modulated bits are transmitted through the optical channel by the LED light. The information is extracted by the receiver camera built-in Laptop used through the capture and processing of frames utilizing an integrated OpenCV library platform. After the finishing of the decrypted assignments, the resulting information is displayed.

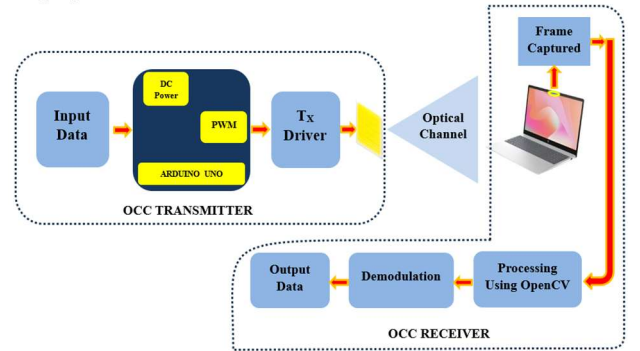


Fig. 3. Proposed OCC system block diagram.

A. OCC Transmitter

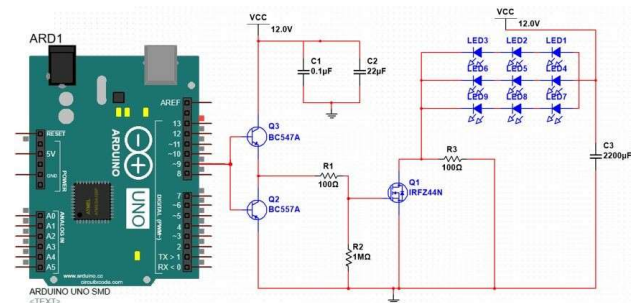


Fig. 4. OCC transmitter schematic.

In Optical Camera Communication (OCC) systems, the transmitter boasts low power exhaustion, rendering it cost-efficient by leveraging pre-existent illumination

hardware, such as Light-Emitting Diodes (LEDs). Furthermore, the circuitry design is uncomplicated, necessitating only a signal modulator and a power source. Fig. 4 presents the schematic diagram of the transmitter circuit, which consists of an Arduino UNO board and the driver circuit used includes an amplification stage by transistors Q2, and Q3 are employed to enhance the voltage present at the microcontroller's output and MOSFET Q1 is employed for current amplification. This circuit presents the benefit of being of a simplistic nature, yet possessing the ability to achieve rapid switching speeds, thereby enabling it to promptly turn on and off.

The transmitting message is text data that is classified into packets. Each packet includes 1 start bit, data frame bits that include the actual data being transmitted, an optional parity bit, and 1 or 2 stop bits. Upon receiving data from the data bus, the transmitting module proceeds to append a start bit, a parity bit, and a stop bit, thereby creating a comprehensive data packet. Subsequently, the data packet is transmitted serially as illustrated in Fig. 5. The data can be encoded using OOK and can be expressed as:

$$s(t) = \sum_{n=0}^{N-1} d_n \cdot \text{rect}\left(\frac{t-n}{T}\right) \quad (4)$$

where d_n is the data bit, T is the duration of each bit, and $\text{rect}(t)$ is a rectangular function that is 1 for $-0.5 < t < 0.5$ and 0 otherwise.

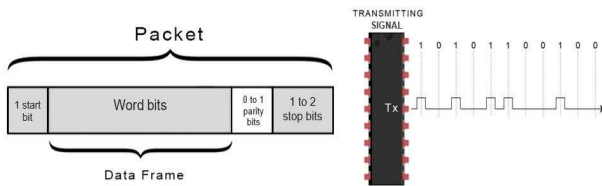


Fig. 5. Transmitting data format.

B. OCC-Based OOK Modulation

OOK is an acronym that denotes On-Off Keying Modulation. This particular modulation technique entails the conversion of logic bits, namely 1 and 0, into on and off-duty cycles. It is widely considered one of the most prevalent approaches for data communication because of its extraordinary simplicity of execution. Pulses utilized in the implementation of On-Off Keying (OOK) modulation for a Visible Light Communication (VLC) system serve to represent the states of light being On and Off. It is therefore apparent that the attainment of OOK modulation is easily achievable through a process of transforming the receipted data to the pertinent on and off-light situations [25, 26]. The procedural execution of OOK Modulation through the utilization of a programmable board is presented in Algorithm 1. In this context, the first function is employed to set an LED Array of interconnected pins and predetermined test information. The second function is utilized for recurring tasks, which encompasses the dispatching of the start bit method to achieve synchronization prior to the transmission of every byte. The sequence is initiated by the transmission of every individual unit of information within a byte. Additionally,

the transmit function encrypts the bits and regulates LED conditions, correspondingly.

Algorithm 1. OOK Modulation

```

1- Function Setup
2- Set pins for an LED Array.
3- Adjust a message.
4- End
5- Repetition
6- Transmit (starter bit)
7- For every data byte
8- For every data bit
9- Transmit (bit)
10- End
11- End
12- End
13- Function Transmit (data)
14- If data is 1 then
15- LED On
16- delay(z)
17- LED Off
18- delay(z)
19- else If data is 0 then
20- LED Off
21- delay(z)
22- LED On
23- delay(z)
24- End if
25- End
    
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C. OCC Receiver

Fig. 6 presents procedures for using a laptop camera as an optical receiver in which captured frames are scanned row by row and a frame-by-frame decoder in the programming language of Python is implemented.

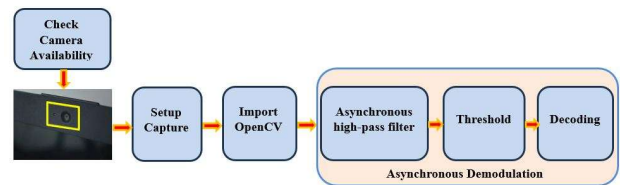


Fig. 6. Procedures of an OCC receiver.

Generally, in the case of an OCC system, it is necessary to construct the system with the aid of OpenCV in order to effectively procedure the captured frame from LED lighting and subsequently extract the relevant data. OpenCV is a comprehensive collection of programming missions that primarily specializes in the realm of real-time computer vision, and is the ideal tool for this task. Initially, it is imperative to set up the characteristics of the camera in order to construct any Optical Camera Communication (OCC). This entails configuring the Capture Setup to optimize camera focus, frame rate, and exposure duration. Furthermore, the OpenCV library can be utilized to process the LED image and extract pertinent information. This platform is equipped with integrated OpenCV functionality, including pixel reading, thresholding, and object detection. To ensure optimal results, the camera speed is carefully held to concentrate exclusively on the LED Array and expose only the region of interest. Once image exposure and information extraction are complete, the transmitted bit stream may be

successfully extracted and displayed. The detection model for LED pixels is presented as follows.

1) Detection and asynchronous

The detection model reveals the LED pixels utilizing blob detection procedures. Eq. (5) is utilized to calculate a pixel as expressed.

$$I_{(u,v)} = \frac{events}{B + |\sum \sigma_k|} \quad (5)$$

where σ_k represents poles for each event taking into account the $B > 0$, to ensure that the result remains bounded. Raised indicators signify a light exporter that flashes repeatedly producing several events with stable positive and negative polarity, which is common in communication channels. A threshold is set to eliminate low-value pixels, and an image morphology technique is utilized to produce binary images to occlude blobs inwards the images and enhance detection.

We combine event information using an event-based asynchronous high-pass filter that eliminates low-frequency noise from event information while maintaining high-frequency information and the high-pass filter equation as expressed in Eq. (6):

$$L_p(t) = e_p(t) - \alpha L_p(t) \quad (6)$$

where $e_p(t)$ represents the event flow at pixel p , while $L_p(t)$ denotes the identical continued time density evaluation. The parameter α is the preset frequency that influences the rate of density drop. However, it's essential to recognize that this equation is processed asynchronously, with the solution being calculated only when an event occurs. Similarly, the limit is evaluated asynchronously, but only during events that reduce the arithmetical workload.

2) Decoding

The time amidst the start and end bits is segmented into bit slots, dividing each data packet into binary bit numbers. The bit values are determined by analyzing the time setoff of the specific high and drop edges of the series in the forward direction. Initially, the parity bits are verified, and if they are accurate, the bits are then decoded. The decoding process can involve thresholding and segmentation, represented as:

$$D = f(I_{(u,v)}) \quad (7)$$

where D is the decoded data, and $f(\cdot)$ represents a function that processes image data to extract transmitted information.

V. EXPERIMENTAL SETUP

To assess the efficacy of the proposed LEDs in a Laptop camera system based on OCC, we build an experimental of the OCC prototype as illustrated in Fig. 7. The input message we transferred is "optical", this is converted to binary using the ASCII encoding scheme. It was sent via a serial monitor from the Arduino Studio application installed on a mobile phone to an open-source electronics

prototyping platform (Arduino UNO R3) using USB-OTG, the LED Array driving circuitry for signal amplification and management of switching RoHS LED Array which was used as an optical transmitter. Its ability to radiate an illuminance intensity of 300 lux at a communication distance of 1.1 meters. We used an Arduino microcontroller that generates a square wave control voltage at 3 kHz. We have opted for the OOK modulation technique owing to its simplicity, but still maintaining a high degree of generality. We used the HP Laptop camera (640 × 480 pixels) as a receiver to apprehend LED agleam with a FOV of 15°. OpenCV library with blob detection technique is imported which prepares the camera to capture frames and decode them in real time. All essential parameters required for the successful execution and implementation of the experiment are listed in Table IV.

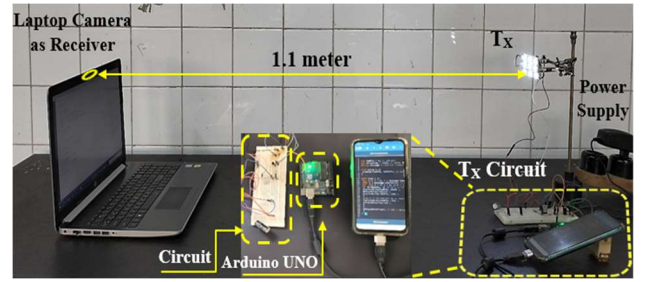


Fig. 7. Experimental Setup for Proposed OCC System.

TABLE IV. OCC SYSTEM PARAMETERS

Parameter	Value
Transmitter Device	12 VDC white LED Array 3 × 3
Microcontroller	ATMega328p
LED FOV [deg]	30
Power [W]	1.5
Resolution	1920 × 1080
Exposure time [μs]	125
Frame Rate [FPS]	25
Ambient light [lux]	100

VI. RESULT AND ANALYSIS

In this section, we present the experiential findings yielded by the proposed OCC system to validate the efficacy of the system.

A. Experimental Results

The system based on OCC is bifurcated into two principal, primarily the transmitter and the other part Receiver. At the transmitter, the message is blinked through the Arduino software with a speed (9600) baud rate via LED Array used using on-off-keying modulation. The aim of using an LED Array is to increase the area of the emitter, thus the dropping of it in the camera, and achieve an indoor environment lighting. At the receiver, a default laptop camera is employed to capture the incoming light signals. The camera sensor acts as the light detector and converts the received optical signals into electrical signals. It can distinguish between high-frequency flashing LEDs and other common events by comparing the voltage controlled by the microcontroller and the data recorded by the camera. We used a digital storage oscilloscope to assign the output voltage of the microcontroller. Fig. 8 illustrates that the incorporated data corresponds to the

control voltage in the form, which expounds the camera's capability to capture data and high-frequency LED lighting with high precision.

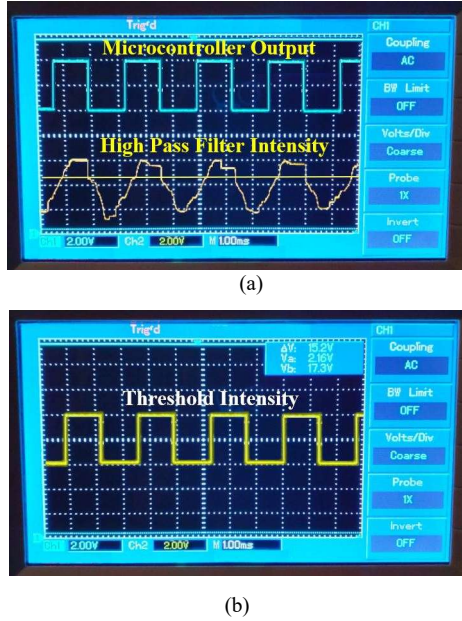


Fig. 8. Control voltage and high-pass filtered for the proposed OCC system; (a) Microcontroller and filter outputs; (b) Threshold Intensity.

By importing the OpenCV library, the laptop camera is initialized to sense the streaming LED light states through Python code, and significant parameters of an OCC system used are listed in Table IV. Fig. 9 shows the captured LED array and their pulses are sensed by the camera in real-time.

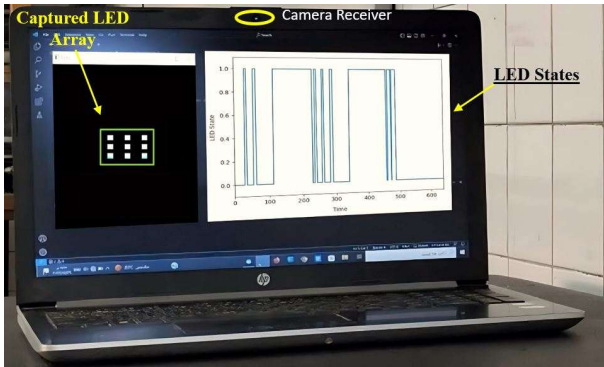


Fig. 9. Received LED array light states by laptop camera.

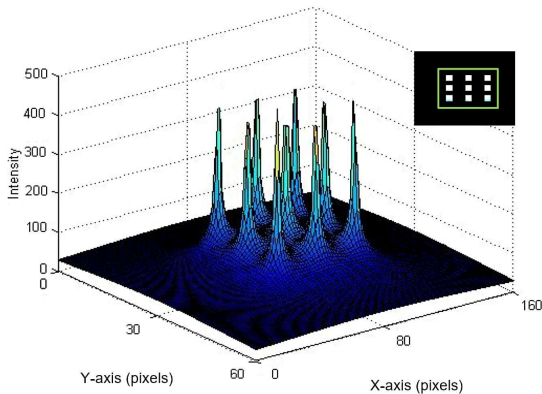


Fig. 10. Intensity profile of captured LED array.

The camera sensor converts the incoming light into an image. Image formation consists of an array of photosensitive pixels that capture the light intensity at different positions on the sensor's surface. The intensity values from the pixels form an image. Fig. 10 plots a 160×60 -pixel size to generate the corresponding image intensity of the captured LED array by the laptop camera used with a speed of $1/6000$ s for a 1.1-meter OCC connection which can distinguish pixel intensities (ranging from 0 to 500) of every LED peak readily.

For this practical system, the data transfer rate meets the requirements, in addition to opting for OOK modulation providing a data rate of half the blinking and camera frame rate. Also, the message transmission time doesn't exceed a few seconds for a message consisting of six or fewer ASCII characters and the LED must possess the ability to be scanned from a practical distance. It is noteworthy that the further the camera is from the LED, the higher the probability of interference with frames. Fig. 11 presents the transmitted message as ASCII data and the received message in binary then converted to generate the ASCII value. The generated ASCII data feeds to the receiver end-user interface to display.

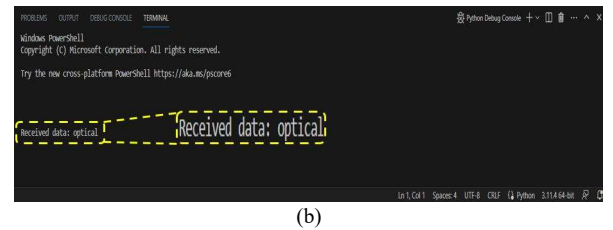
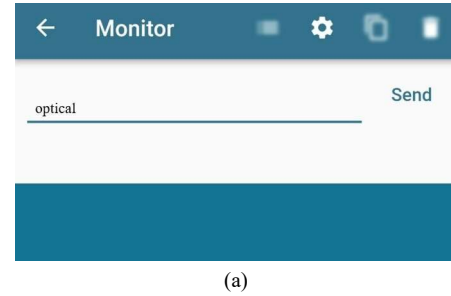


Fig. 11. Interface (a) transmitted message (b) received data.

B. System Performance

In this section, we present some system performance findings obtained from the proposed OCC system for the transmitter and receiver. This intends to substantiate the efficiency of the proposed system and provide valuable insights for augmenting the performance of practical OCC.

C. A. LEDs vs. Transmission Power and BER

The number of LEDs refers to the quantity of light sources used to transmit the optical signals. Each LED can emit a certain amount of light, and multiple LEDs can be used together to increase the overall optical power. In our experiment, we used a white LED Array of 9 LEDs. The optical transmission power and the number of LEDs are interrelated increases in the overall transmission power as shown in Fig. 12(a). This result confirms the scalability of

the system, where increasing the number of LEDs can effectively enhance the overall transmission power.

As the transmission power increases, the received Signal Strength (RSS) also increases with the number of LEDs as shown in Fig. 12(b). This indicates that the receiver can effectively capture the enhanced optical signal as more LEDs are employed. The increase in RSS is crucial for improving the system's performance, as a stronger signal facilitates more accurate detection and demodulation at the receiver.

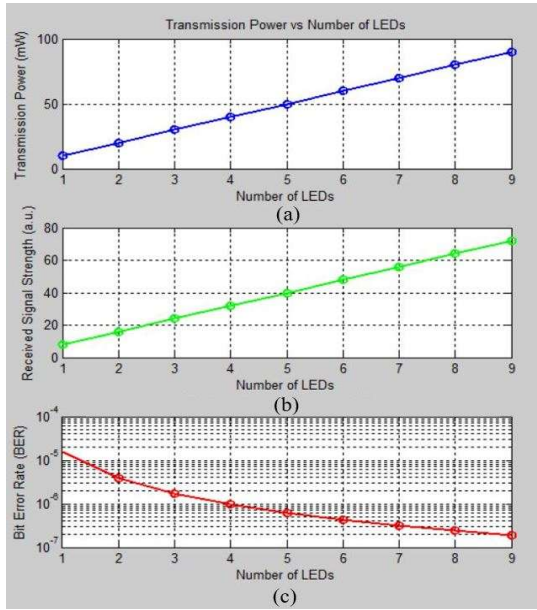


Fig. 12. The efficiency of (a) transmission power, (b) received Signal strength, and (c) BER.

In Fig. 12(c), The BER decreases significantly as the number of LEDs increases. This reduction in BER is attributed to the higher received signal strength, which improves the Signal-to-Noise Ratio (SNR) at the receiver. A lower BER is essential for achieving higher data rates and more reliable communication. The results demonstrate that increasing the number of LEDs can effectively mitigate errors in the transmission, enhancing the overall quality of the communication link. The results of our system using more LEDs turned out to have a relatively better BER performance.

Fig. 13 shows the transmission rate of optical signals and the connection distance. From this figure, the longer the communication distance, the lower the required transmission rate at a shorter distance of 20 cm or more, our system is capable of transmitting signals exceeding 1 kbps with a data precision rate of over 95% and a data bit precision rate of just over 97%. The data precision was evaluated based on the ratio of correctly received data bits to the total number of transmitted bits during the experiment.

Even with RoHS LEDs, the system realizes 100% transmission precision at data transfer rates as low as 32 Kbps at a transmission distance of 1.1 meters. The system evaluates the precision of 8-bit ASCII decoded characters and demands complete precision for every character. The successful decoding process ignores the start and end bits,

focusing instead on the accuracy of all binary bits through the automatic decoding repetition rate. The SNR at the receiver was measured to ensure that the received signal was sufficiently strong to achieve high precision. Consequently, as the SNR improves the BER for the information message rapidly decreases to zero, while the bit BER declines more gradually to 0.5.

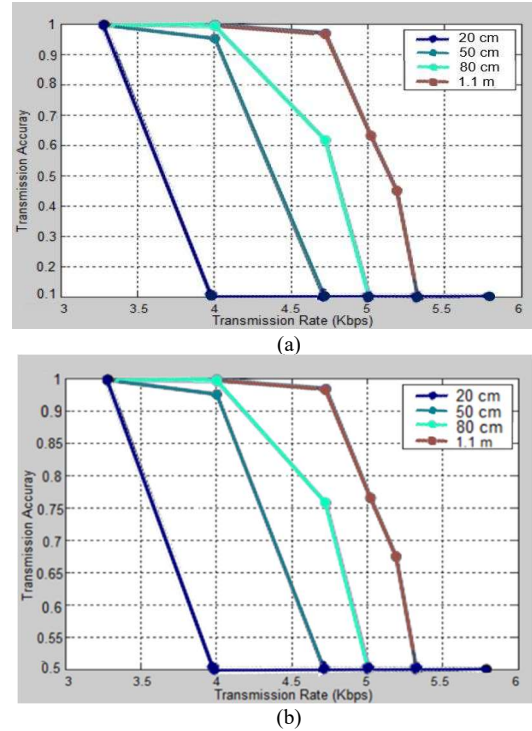


Fig. 13. Transmission rate versus distance for (a) data accuracy and (b) bit accuracy.

D. Exposure Time vs. BER

Fig. 14 illustrates the relationship between Bit Error Rate (BER) and exposure time in an OCC system under varying transmission distances. The results demonstrate a clear dependency of BER on both exposure time and distance. At shorter distances (0.2 m, 0.5 m, and 0.8 m), the BER remains consistently low, approaching zero across the tested range of exposure times (50 μ s to 200 μ s). This indicates robust communication with minimal errors under these conditions. However, at the longest tested distance of 1.1 m, the BER exhibits a more pronounced dependence on exposure time. Specifically, at shorter exposure times, the BER is significantly higher. For instance, at 50 μ s exposure time, the BER at 1.1 m is approximately 0.033. As the exposure time increases, the BER at 1.1 m decreases. Notably, at the target exposure time of 125 μ s at 1.1 m, the BER is reduced to approximately 2.02×10^{-3} , this reduction in BER with increasing exposure time at longer distances can be attributed to the increased integration of received light by the camera sensor, leading to a higher signal-to-noise ratio and consequently fewer errors in demodulation.

The results underscore the importance of optimizing exposure time based on the communication distance in OCC systems to achieve acceptable BER performance.

While shorter distances allow for a wider range of exposure times without significantly impacting BER, longer distances necessitate careful selection of exposure time to mitigate the effects of signal attenuation and noise. These findings suggest that for reliable communication at 1.1 m, an exposure time of at least 125 μ s is desirable.

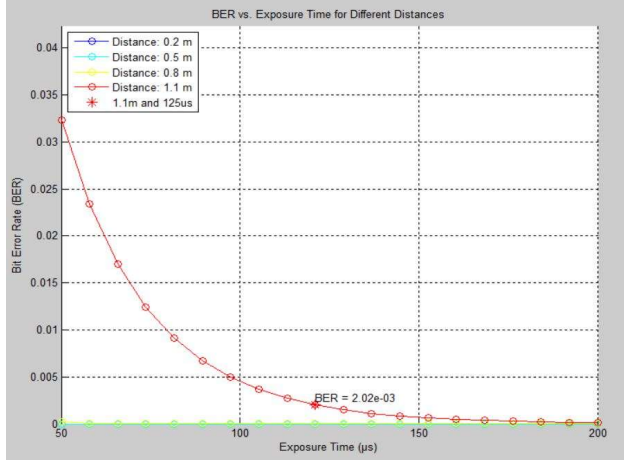


Fig. 14. The exposure effect of an OCC system.

E. Comparison with Existing OCC Systems

Table V presents a comparative analysis of the proposed method against existing modulation techniques utilizing LED arrays as transmitters. The table highlights the unique characteristics of the proposed OOK modulation, which includes point-to-point transmission and unidirectional communication capabilities. Unlike other methods, the proposed approach is compatible with various camera types, offering flexibility in deployment. The communication distance of the proposed method falls within the medium range, up to 1.1 meters, while achieving a higher data rate compared to other studies. Furthermore, the proposed OCC system demonstrates superior performance in terms of Bit Error Rate (BER), outperforming the work in [13]. The low BER value signifies minimal data loss during transmission when using a 3×3 RoHS LED array, indicating the effectiveness of the proposed method in ensuring reliable communication.

TABLE V. COMPARISON OF PROPOSED VS. EXISTING TECHNIQUE

Parameter	Ref. [13]	Ref. [14]	Proposed
Modulation	OOK	OOK	OOK
Camera type	Rolling shutter	Both types	Both types
Distance	1.4 m	20 m	1.1 m
Point-to-point	Not mentioned	Not mentioned	Yes
Data Rate (Kbps)	13.44	7.680	32
BER	$> 10^{-2}$	$< 10^{-4}$	$> 10^{-2}$

The proposed Optical Camera Communication (OCC) system offers a compelling alternative to traditional Radio Frequency (RF) communication systems, particularly in terms of speed and accuracy. While RF technology has been widely adopted, it often faces challenges related to

interference, limited bandwidth, and security concerns. In contrast, OCC leverages the advantages of optical communication, enabling high-speed data transmission with minimal latency. The system's ability to utilize existing camera infrastructure and its inherent resistance to electromagnetic interference make it an attractive option for various applications. We aim to highlight the superior performance of OCC as shown in Table VI.

TABLE VI. COMPARISON OF OCC AND RF

Parameter	OCC	RF
Data Transmission	Using light signals, specifically from LEDs, which can transmit data at much higher frequencies. This allows us to achieve high data rates.	Traditional RF systems typically operate in the range of a few kilobits to several gigabits per second.
Spectrum Availability	The optical spectrum is vast and largely less susceptible to	Limited and crowded.
Interference	electromagnetic interference	High
Accuracy	High	Low

VII. DISCUSSION

The performance of our OCC system's data rate is highly contingent on the precision of the chosen emitter and receiver. On the emitter side, the light's switching speed determines the maximum rate at which information can be transmitted. On the receiver side, the frame rate and resolution of the camera impact its ability to perceive variations in light. Ideally, a perfect camera would align each pixel line of an image to distinct light states during the frame time, achieving optimal data transmission rates.

In practical terms, off-the-shelf cameras lack this precision, necessitating compromises. The system benefits from representing each state of the light with well-defined. The decoding speed then becomes proportional to the ability for accurate data recognition and the overall height of the camera's image. Moreover, while leveraging high-frequency modulation to transmit data can mitigate flickering, careful consideration is required to balance between the data rate and the human perception threshold. Insufficiently high modulation frequencies can produce noticeable flicker, causing discomfort or health issues. In our implementation, we used a modulation frequency of 3 kHz, which is above the Critical Fusion Frequency (CFF) of human vision, typically around 60–100 Hz, to reduce flicker. Nonetheless, we acknowledge a minor percentage of flicker, evident in our results, which can be further addressed in future work.

Furthermore, regulatory limits and the inherent physical characteristics of LEDs constrain the available frequency spectrum for modulation. This constraint directly impacts achievable data rates, potentially limiting the system's throughput compared to RF technologies. Additionally, the range and achievable data rates are also affected by factors such as the power of the LEDs, the sensitivity of the camera, and the distance between the transmitter and receiver. Signal attenuation over distance requires either higher LED power or more sensitive

cameras to maintain acceptable BER, complicating hardware dependencies further.

VIII. CONCLUSION

This paper has significant theoretical and practical implications. Theoretically, the development of an Optical Camera Communication (OCC) system using LEDs and a default laptop camera enriches the existing body of knowledge on wireless communications. The implementation of On-Off Keying (OOK) modulation demonstrates a feasible method for transmitting text data using visible light, which is a novel contribution to the field of both optical communications and the Internet of Things (IoT). Practically, the system showcases the potential for simplifying and enhancing data transmission methods in various real-world applications, particularly where conventional wireless communications face limitations.

The practical advantages of this research are manifold. One of the most prominent benefits is the potential for using inexpensive and readily available components, such as LEDs and default laptop cameras, to achieve efficient wireless communication. This makes the technology accessible for widespread use in various applications, such as smart homes, healthcare monitoring, and industrial IoT systems. The system's ability to function with low flicker and high transmission rates also makes it suitable for usage in environments where minimizing visual disruptions is critical.

Despite its promising results, this research has several limitations. One of the primary limitations is the relatively short transmission distance achieved in the experiments, which was capped at 1.1 meters. This restricts the system's applicability in scenarios requiring longer-distance communication. Additionally, the dependency on ambient light conditions and the fixed frame rate of the receiver camera (25 fps) could limit the system's performance in various lighting environments. Future enhancements could address these limitations by optimizing the modulation techniques and improving the sensitivity of the receiver camera.

To build upon the findings of this research, future studies could explore the following areas:

To further improve and develop applications based on optical camera communications, in the transmission process to prevent the flickering of the continuous light source, we suggest using a modulation technique that employs a combination of Frequency Shift Keying (FSK) and On-Off Keying (OOK) modulation techniques. FSK will be utilized for flicker mitigation in scenarios requiring low-frequency data transmission, while OOK will be employed for high-frequency data transmission for OCC and VLC systems. For the receiver, when the camera frame rate becomes 240 FPS and more in intelligent devices, the experiment can be iterated to determine the viability of the technology.

Energy Efficiency and Power Management: Using energy-efficient algorithms and power management techniques that can be integrated into OCC systems, which

would be critical for battery-operated devices and sustainable applications.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

M.A.R.E suggested the research problem; M.A.H designed the proposed system and wrote the paper; A.H analyzed the data; H.A.H reviewed and supervised the paper; T.H added the result analysis; A.M.E-R provided extensive feedback, provided edit review and proofread the paper; all authors had approved the final version.

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